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2026 ??????

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(???? 18 ?????? 500) ???????? ????????? ???????

????????	95% ????	???????	??????
$t(499) = 33.93$, $p < 0.001$, $d_z = 1.52$	$[0.779, 0.875]$	$+0.827$	ΔSR$_{\text{eval}}$
$t(499) = 33.74$, $p < 0.001$, $d_z = 1.51$	$[10.64, 11.95]$	$+11.30$	ΔSR$_{\text{comp}}$
$t(499) = -74.41$, $p < 0.001$, $d_z = -3.33$	$[-0.188, -0.178]$	-0.183	Δ engagement
??????? ?????? ??? ?????? ????	HDI $[0.7793, 0.8764]$	$+0.8265$	Bayesian ΔSR$_{\text{eval}}$

?? ??????? ??????? ????? .???? ???? ?? ??????? ?? ??????? ??????? ????? ?(0.6265+ ????? ???) ??????? ???? ???? (B) ???????? ?????? ???? ?(0.0121-) ??????? ????? ?? ?????? ERIC ????? ?(0.1905+) ???? ????? BRS ??? ?????? ??? ?? ???? 256 ??? ?? ???? ???? .0.3791+ ????? B$\times$C ???????? ?? 78.5 ?????? ??? ????????? ??????? ??? .(PRCC $+0.77$) cred_dbi ????????? ????? ??? ?(,0.383] [0.802 ??????? ????? ?0.567+ ??????) ?????? ????? ??? .???????? ??????? ????????? ??????? ???? ??????

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??????? ?????? 52??? Respondents .(0.827+ ????? 0.808+) ????????? ?? ???????? ?????? ?????? ?????????? ?????? ??????? ??? ????????? ?????? ?????? ????? (Gemini) ???? ????? ???????? ????????? ????? ????? .??????? ??? ?(0.163+) ??????? ????? ????? (2.650+) ???? ????? ????? ???????? .????? ?????? ????? ???? 18 ?????? ????? ?????? ???? ?????

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1. ?(EST-P1) ??????? ?????? ???????? ????????? ?????? ????? ????? ?? .

2. ?(EST-P2) ??????? ??????? ???????? ?????? ????? ????? ?? .

3. ?(EST-P3) ?????? ????????? ?????? ???????? ??????? ???? ?? .

4. ?(EST-P4) ?????? ???? ????? ?? EST ????? ?????? ?? .

5. ????????? ??????? ???? ???????? ???????????? ?? .

6. ????????? ??? ?????? ??????? ??????????? ??????? ???? ???? .

:???? ??? .?????? ??????? ????? ?????? ?????? ????? ?????? ???? ???????? BRS, MORE, ERIC, Echo Meter,) ??????? LOKLOK ????? ???????? ?????? EST ??????? ?????? ????? ???? ????? ????? ?(Algorithmic Cooling, Zewail gate ????? .????? ????? ?????? ?GRM-IRT ???? ????? ???????? ????? ?????? ???????? ???? ????????? ?? ?? ??? BGE-M3 ??????? ????? ???????? .?????? ???????? ?????? ?? ???? ??????? ?????? ?????????

????????? EST ??????? .3

??? ????????? ?????? :EST-P1 .??????? ??????? ????? EST ????? .??????? ??????? ??? ????? ???????? ?????? :EST-P2 .????????? ????? ???????? ?? ???????? ?????? :EST-P4 .???????? ??? ?????? :EST-P3 ???????? H3 ??????? ???? H2 ????????? ????????? H1 :????????? .???????? ??????? H6 ?(?????????) DeGroot ?????? H5 ???????? H4 ???????????? .(?????????) ???????? ?????? H7 ?(?????????) ??????

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????? ????? ??????? PB ??? ????? ???????? ???? ????????? ???????? ?? BRS ???? ?BRS? ????? ?????? ?????? ????? ??????? ??????? ?? ????????? ????????		H1, H4, H5	EST-P1
???? ????? ?????? ?? MORE/Lagrange ???? ?????? ???????? ?Echo? ??????? ????? ??????? ???????? ??? ????????? ?? ????? ??????????? ??? ?????? ??????		H2 ??????	EST-P2
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21 ?? ??????? ??? ???? .????? ????????? 60?? ????????? ?????? ???? ?????? ?????? ????? .??????? ????? 52 ???? ?????? ?????? ?????? ????? ???? ?? (PB) ??????? ??????? ??????? ?????? 11 ?? (BRS) ?????? ????? ??????? .?????? ????? ???????? ?????? ??????? ???? ??????? ?????? ????? ??????? ??? ?($\omega = .964/.962$?PB $\alpha = .947$?BRS $\alpha = .957$) ????? ??????? ????? ??? ?????????? ?????? ??? ???????? ????? .???? ??????? ??????? .??????? ?????? ??????? ????

185 ?? ???? ????? ??? ??????? 160 ?? ????? ?????? ?? ?????? ?????? ???? .????????? ?????? 25? ???? ????? 4,828? ??????? ?????? ???? 60? ??????? ??????? ???? 167 ???????? ?????? ????? ????? ??? 31 52??? ??????? ???? .??? 198

???????? Sentence-Transformer ???? 4.1

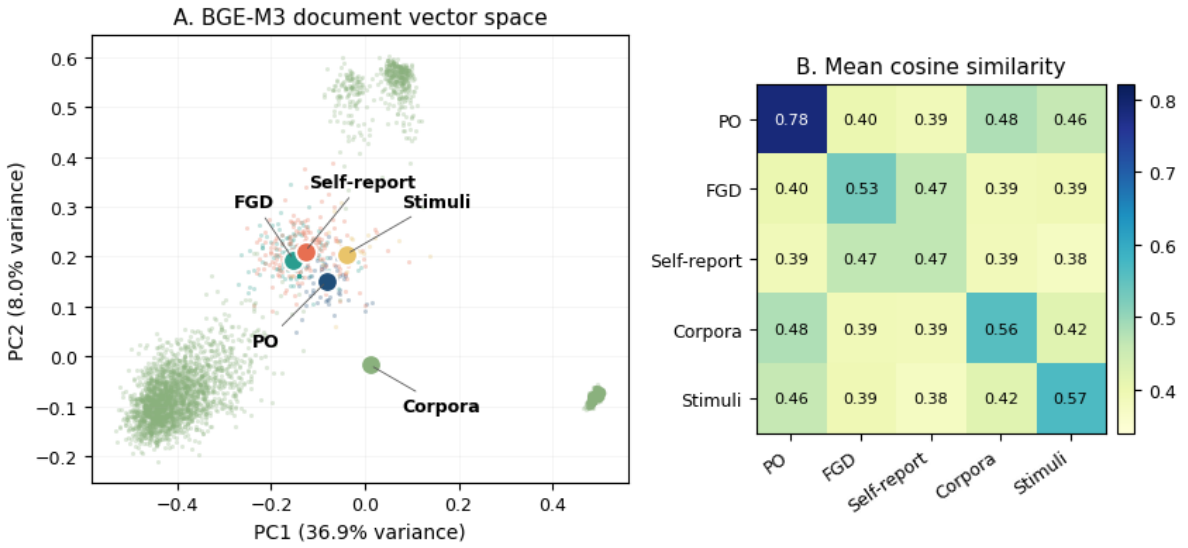
??? ????? ?????? ????? ??? ????? ??? ?BAAI/bge-m3 ?? ??????? ??????? ??? ?????? ????? 1,024 ?????? ?????? ?????? ???? .XLM-RoBERTa ???? ??????? ????? ?????? ??? ???? ?????? ????? ???????? ?????? .????? 8,192 ????? ?? (zero-shot): ????? ????? ??? ?????? ??????? .???????? ????? ?? .??????? ?? ?? ????? ????? ?? ???? ??? ????????? ?????? ???

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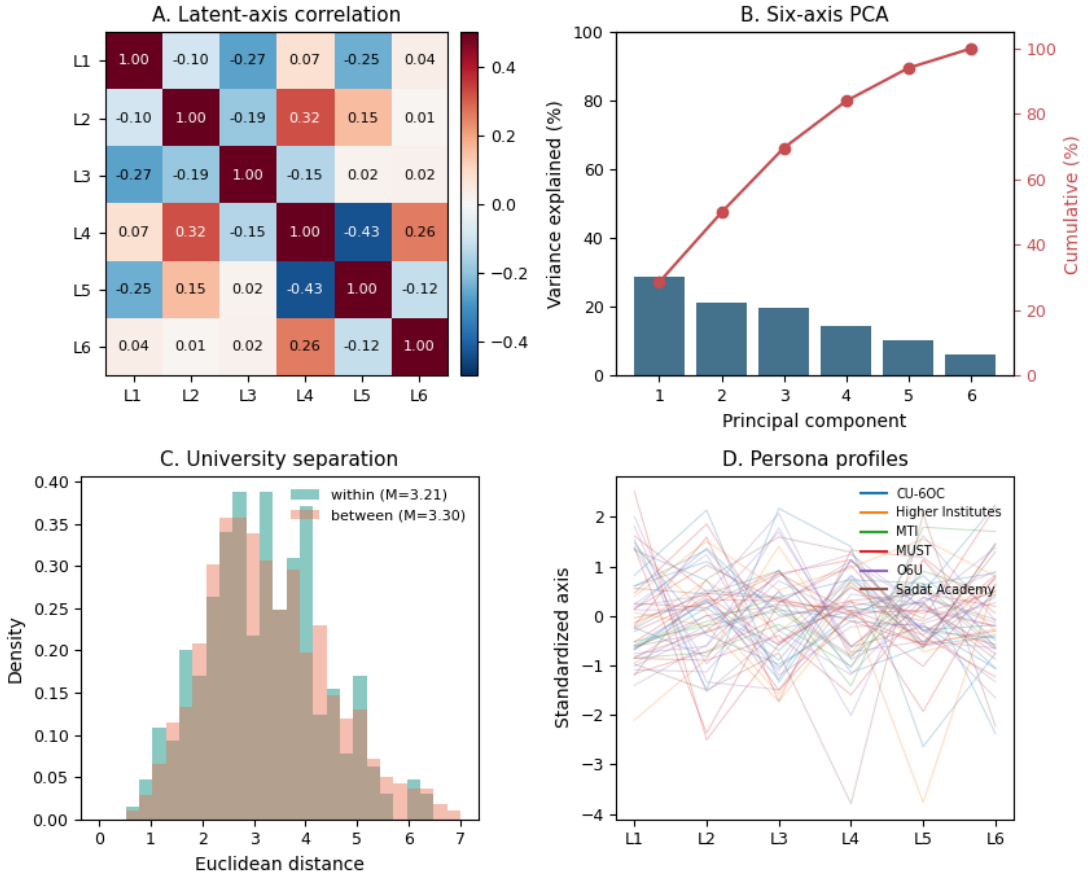
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??????? ?? 44.9% ??????? ???????? ???? .????? ????? 5,258 ???????? ?(0.785) ????????? ???????? ?? ?????? ???? ???? ????? ???? .(8.0%? 36.9%) ???????? ?(0.531) ??????? ??????? ?(0.559) ?????? ?(0.568) ???????? ?? ?????? ?????? ????? ?(0.48–0.38) ??? ??????? ??? ??????? .(0.466) ?????? .???? ?? ????? ??? :0.196 ???????

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??????? .(?) ?????? ??????? ??????? (?) BGE-M3 ????? ?????? ???? :1 ????? .????? ????? ????? ???? ?????



????? ?(?) ??????? ???????? :????? 52??? ?????? ?????? ??????? :2 ????? .(?) ????????? ???????? ?(?) ???????? ???/???? ???????? ?(?) PCA

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PCA + LLE) ?????? ?????? ?? ?IPW? DML?? ?????? ????? ???????? ?? ????? ?DML ??? 0.042 ??? 0.887 ?? ???? ?????? ???? ????? .(+ Laplacian eigenmaps ??? ??? .?????? ?? ????? ??? ??????? ?(0.127) ???????? ??? IPW ????? ??? ????? ????? ???? ??) ????? KDE ????? .4.82 ??? ??????? ????? ??????? .????? 500 ?????? ?Q_0 ??????? (?????? ??? ??????? ???

???? ????? .???? 18 ??????? $2^3 = 8$???????? ??????? ??? ????? ?? ??????? Hamiltonian, Laplace, Lyapunov, DeGroot, Lagrange, :?????? ?????? ??? ?????? ??? ?????? ????? ???????? ???? ????? ????? .Hilbert, Percolation, Entropy ?? ????? 35?? Samejima GRM-IRT ?????? ???????? ?????? ?????? .???????? ?5.119 ?????? PB ????? ??? G0 ????? ??????? .MCS+PB ?? ???? 8? NMLS .52??? ?? ??????? ????? ?????? ????? ?????? ?????? ??????

????????? t ???????? ?????? ???????? ??????? .??????? ????????? ?????? ??????? ??????? ???????? ?95% ???????? ????? ?????? ?Cohen's d_z? ?????? ???? ????? ????? ????? ???? ??????? ???????? .Wasserstein ??????? ?????? ???????? ?????? ???? ???????? ???? ????? ????? ??? ????????? ?$P(\Delta > 0)$?????? ?? ????? 256 ???? .??????? ???????? ??? ??? ????? ??? .????? ?? ?????? ??? ???? ??

.????? 192 = ???? 8 $\times$????? ????? 24 ???? ?????? ????????? ????? ??????? ???? ?? ?????? .???? 18 ?????? ????? ?????? ???? ????? ???? ????? ??? .?????? ???? ???? ?? ??????? ?????

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?7–1 ????? ??? 0.827+ ?????? ???? :????? ?????? ?? ???????? ??????? .1–0 ????? ??? 0.183- ????? ??????? ?100–0 ????? ??? 11.30+ ????? ????? .($d_z = -3.33$) ?????????? ????? ??????? ?????

A$\times$C ?A$\times$B $+0.0216$?C -0.0121 ?B $+0.6265$?A $+0.1905$:??????? ??????? ????? ????? BRS ????? ?????? MORE .A$\times$B$\times$C $+0.0886$?B$\times$C $+0.3791$?-0.0084

??????? ??? ????? B×C ???? .???????? ?????? ??? ?????? ???? ERIC? .??????? ??? ????? ??????? ????? ???? ??? ??????

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??????? ????? ??? Hilbert? Lagrange? DeGroot ?? ???????? ????? ????? ????? ,0.197] 0.233 ?????? $\triangle SR_{eval}$???? DeGroot ????? .???????? ??? ?? ??????? ?????? Hilbert ?????? ?[,0.713 0.787] 0.750 ?????? Lagrange ?????? ?[0.269 ????? ??? ?B ?????? ?????? ??????? ?? Lagrange .[,0.402 0.503] 0.453 ???? ??? ??????? ?????? ???? ??????? ???? ???? ????? ?? Lagrange-off .???? Lyapunov? ?????? ??? ??????? ???? Entropy? Percolation? Laplace .????? .??????? ???? ??? ?????? Hamiltonian?

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??????-???? ????? ????? ?????????? ?????? ??? ????? ???? 256 ??? ?0.593+ ???????) 1.810+ ??? 0.244- ?? ??????? ?????? .????? ???? ??? ???? .????? ???????? ?? 96.5% ?(,0.383] [0.802 ??????? ????? ?0.567+ ?????? cred_dbi :??????? ????? ???????? ???? .78.5 ?????? ??? 0.827+ ???????? ???????? echo?? ??????? ?????? .0.27- cred_echo ?0.31+ cred_brs ?0.77+ ?????? ??? ???? ?????? ????? :????? ????? .???? ???? ????? :??????? ??? .??????? ????????? ???? ?????? ?? ??????? ???????? ?????? ???? ???????? ??? ?????? ????? ????????? ??????? ???? ??????? ??????? ?? .????? ???? ?????????

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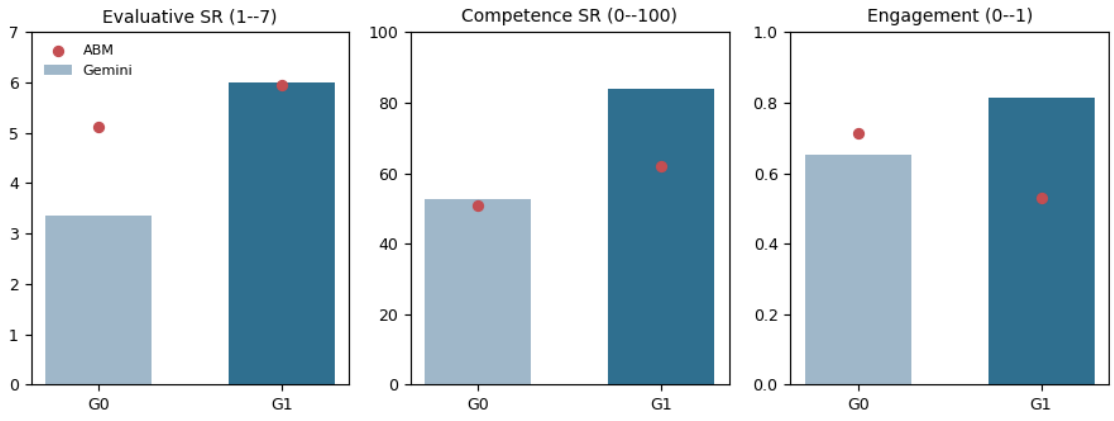
50??? ?(0.827+ ????? [,0.760 0.856] 0.808+) ???? 60 ???? ????? ??????? ?(???? 100% ?[,0.805 0.884] 95% ?0.839+ ???????) ??????? ?????? ????? ???? ??????????? ????? .(0.832+/0.826+/0.827+) ER/WS/BA ???????????

B ???? .??? ???? A? ?BRS ?????? ??? ??? ???? ??????? ????? :?????? ????? ???? ??? ?????? ???? ??????????? ???? ??????? ?? ????? ?????? .?????? ?????

????? ??????????? .??? 198? ???? 52 ?????? ?????? ???? ?????? ?????? ?583. ??? R^2 ???? ?????? ??? 167 ?????? ?($R^2 = .915$) ???? PB?? ????? ???????? ?????? ????? ?? ??????? ??? ?? .???? R^2 ?? ??? ??????? ???????? ????? ??????? ??? ?? ????? ??? ???? 167? ???? 52 ??? .?????? ???? ?????? ??????? ??? ?????? .???? ???? ?? ????? ??? ??????? ?? ??? ?????? ???? ??? ????? ??? ?1.69 ???? ??????? ???? ????? ?? 0.763 ???? ????? ??? ???? ?????? ?0.636+ ??? ΔSR ????? .?????? ????? ????? ?????? ???? ???????? .????? ??????? ????? ?[,0.261- 0.122-] 0.191- ???????

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[,2.342 2.967] 2.650+) ?????? ????? ???????? ?????? ????? ??????? ????? ????? ????? ????? ?(11.30+ ????? 31.15+) ??????? ???? ?????? ?(0.827+ ????? MAE :SR_{eval}?? ??????? ??????? ?????? ????? .(0.183- ????? 0.163+) ??????? ?????? ??????? .Spearman $\rho = 0.071$?Pearson $r = 0.269$?RMSE 0.842 ?0.655 ?? ????? ????? ?? ?????? ??????? ????? ?? ???? ?????? ?????? ??? ??? ?????? ????? ???????? .??????? ????? ???????? ???? ???? ????????? ????? .???? 18 ?????? ????? ????? ?????? ?????



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?????? ?????? 52??? ?????? .?????? ?????? ????? ???????? ????? ?? NMLS ??? .?????? 500 ????? ??????? ????? 500??? ???????? .??????? ???? ???? ????? Hamiltonian .????? ??????? ???? ??? ???? ????? ????? ????????? ???? ???? ????? ????? BRS .????? ?? ??????? ?????????? ??????? ???? ??????? .v8 ???????? ????? ??? ????????? ????? ?????? .????? ???? ????? .??????? ??????? ??????? ??????? ????????

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?? ??? ??????? ??????? ???? ????????? ?????? ?? ?? ??????? ????????? ??????? ??????? LOKLOK ????? ???????? ????? ??? EST ???? .???? ????? ???????? ????????? ??????? ????? ???????? ????? ??????? ?????? ????? ????????? ??????? ?????? ?? ???????? ???????? .?????? ??????? ????? ?? ???? EST ??????? ????? .??????? ??????? ???? ?????? ??? ?????? ????? .????????? ????? ???? ?????? ?????? ???????? ???????

?????? 0.183-? ?????? ???? 0.827+ ?? ????? ???? ??????? ??????? ????????? (?????) :???????? ??????? ???????? .??????? ????? ?? ????? ?????? 96.5%? ?? ???? ???? ?MORE ???? ??????? ??? ?????? ???? ???? Lagrange ????? ??? (??????) .????????? ??? ?? ?????? ??? ???? ?????? :??????? ????? ?????? ??????? ???? ????????? ???? ????? ?????? ??????? ???? ???? ?????? ??? ?????????? ??????? ??? ????? (??????) .??????? ?? ????????? ???? ????? (??????) .??? ???? ?????? ?????? ????? ??? ????? ?????? ?? 78 ???? ?????? ???? ????? ?????? ?? ??? 15% ??? ?????? ???????? ??????? .???????? ??????? ???? ?????? ??????? ????? ????